

Exercise Sheet 11

Predicate Logic

Note that question 2 is marked as being assessed.

Consider the following signature:

- Function symbols: **zero** (arity 0); **succ** (arity 1)
- Predicate symbols: $<$ (arity 2); $\leq$ (arity 2)

We will use infix notation for the binary symbols $<$ and $\leq$. For simplicity we write 0 for **zero**, 1 for **succ(zero)**, 2 for **succ(succ(zero))**, etc. Consider the following formulas that capture properties of the above symbols:

- let S_1 be $\forall x. \neg 0 \leq x$
- let S_2 be $\forall x. \forall y. x < y \rightarrow x \leq \text{succ}(y)$
- let S_3 be $\neg \exists x. x < 0$
- let S_4 be $\forall x. \forall y. \text{succ}(x) \leq y \rightarrow x < y$

1. Provide a constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow \forall x. \neg 0 < x$
2. **assessed:** Provide a constructive Natural Deduction proof of $(S_3) \rightarrow (S_4) \rightarrow \forall x. \neg \text{succ}(x) \leq 0$
3. Provide a constructive Natural Deduction proof of $(S_3) \rightarrow (S_4) \rightarrow \neg \exists x. \text{succ}(x) \leq 0$
4. Provide a model M_1 such that $\models_{M_1} \exists x. \exists y. x < y \wedge \neg x \leq y$
5. Provide a model M_2 such that $\models_{M_2} \neg \exists x. \exists y. x < y \wedge \neg x \leq y$